International Journal of Medical Informatics 191 (2024) 105558

Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf

Telehealth development in the WHO European region: Results from a quantitative survey and insights from Norway

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A B S T R A C T

Background: The COVID-19 pandemic sent shock waves through societies, economies, and health systems of Member States in the WHO European Region and beyond. During the pandemic, most countries transitioned from a slow to a rapid adoption of telehealth solutions, to accommodate the public health and social measures introduced to mitigate the spread of the disease. As countries shift to a post-pandemic world, the question remains whether telehealth's importance as a mode of care provision in Europe continues to be significant.

Objective: This paper aims to present, synthesize, and interpret results from the Telehealth Programmes section of the 2022 WHO Survey on Digital Health (2022 WHO/Europe DH Survey). We specifically analyze the implementation and use of teleradiology, telemedicine, and telepsychiatry. Norwegian telehealth experiences will be used to illustrate survey findings, and we discuss some of the relevant barriers and facilitators that impact the use of telehealth services.

Methods: The survey tool was revised from the 2015 WHO Global Survey on eHealth, updated to reflect recent progress and policy priorities. The 2022 WHO/Europe DH Survey was conducted by WHO and circulated to Member States in its European Region from April to October 2022.

Results: The data analysis revealed that teleradiology, telemedicine, and telepsychiatry are the telehealth services most commonly used in the WHO European Region in 2022. Funding remains the most significant barrier to the implementation of telehealth in the Region, followed by infrastructure and capacity/human resources. The survey results highlighted in this study are presented in the following sections: (1) telehealth strategies and financing, (2) telehealth programmes and services offered by Member States of the WHO European Region, (3) barriers to implementing telehealth services, and (4) monitoring and evaluation of telehealth.

Conclusion: Based on WHO's 2022 survey, the use of telehealth in the WHO European Region is on the rise. However, merely having telehealth in place is not sufficient for its successful and sustained use for care provision. Responses also uncovered regional differences and barriers that need to be overcome. Successful implementation and scaling of telehealth requires rethinking the design of health and social care systems to create robust, trustworthy, and person-centred digital health and care services.

1. Introduction

Over the last decade, the sustainability of health systems in Europe has been impacted by multiple socioeconomic drivers that have exerted pressure on traditional health systems and social care delivery models: demographic change and population ageing, changing profiles and burden of disease, increasing costs of healthcare, health workforce sustainability, growing patient demands, and new health security requirements. Digital health services such as telehealth are essential to mitigating the effects of these drivers and actively engaging patients in their care through the use of digital tools. The main aim of telehealth has traditionally been to provide services to people in remote locations with limited access to healthcare services [1,2]. Additionally, it is valuable for marginalized groups and aging populations.

Telehealth can contribute to achieving universal health coverage by improving patients' access to quality and cost-effective health and welfare services, irrespective of where they are. The World Health Organization (WHO) defines telehealth as the delivery of health-care services, where patients and providers are separated by distance [3,4]. The term includes the use of electronic information and telecommunication technologies to support long-distance clinical health care, patient and professional health-related education, health administration, and public health [5,6]. Telehealth can be delivered in several ways [3,7], "Store and forward" consists of storing and sending information remotely, also described as an asynchronous mode of digital health care. "Interactive services" are described as a real-time or synchronous mode of digital

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https://doi.org/10.1016/j.ijmedinf.2024.105558
Received 18 March 2024; Received in revised form 21 June 2024; Accepted 16 July 2024
Available online 17 July 2024

health care. "Remote care or monitoring" enables health workers to provide follow-up care to a person over distance and in real-time, using technologies such as connected medical devices and sensors. Home monitoring, e-consultations (video and text) [8,9], self-management of disease and symptoms, screening and diagnostic tools, artificial intelligence, and new smartphone applications all contribute to the delivery of health services to people in need of care and follow-up, especially those affected by chronic disease [10].

The WHO recognises several applications of telehealth, including telemedicine, where patients transmit information from sensors and monitoring equipment to external monitoring centers; teleradiology, which uses ICT to transmit digital radiological images for diagnosis or consultation; teledermatology, which uses ICT to transmit medical information about skin conditions for diagnosis or consultation; telepathology, which involves transmitting digitized pathological results, such as microscopic images of cells, for diagnosis or consultation; and telepsychiatry, which uses ICT to provide mental health services [3,9].

Prior to the COVID-19 pandemic, the adoption of telehealth was constrained by a range of barriers, most significantly technological and infrastructure challenges, resistance to procedural and workflow changes, and financial and legal barriers [6]. The COVID-19 pandemic sent shock waves through health systems, societies, and economies, emphasizing the importance of digital solutions for resilient, crisis-ready, adaptive, and agile health systems [11]. During the pandemic, reductions in the number of in-person consultations were partly offset by an unprecedented scale-up of telehealth to provide safe clinical services [12]. As countries transition to a post-pandemic world [13], the question remains whether telehealth will maintain its importance as a mode of care provision in Europe.

Several studies have explored the barriers and facilitators of telehealth services. For example, a qualitative study by Wong et al. examined the perceptions of stroke patients and advanced practice nurses regarding sustaining telecare consultations in nurse-led clinics [14]. Another systematic review and *meta*-analysis investigated the effectiveness of nurse-led telehealth interventions for rehabilitation among community-dwelling patients with chronic diseases [15]. These studies highlight the importance of considering user perceptions, organizational factors, and the specific context of telehealth interventions when assessing their implementation and sustainability [16].

In 2022, a regional survey was conducted in the 53 Member States of the WHO European Region (2022 WHO/Europe DH Survey) to examine the status of countries' digital transformation in support of universal health coverage and better health and well-being. The findings are presented in a WHO regional report, "The ongoing journey to commitment and transformation: digital health in the WHO European Region, 2023" [3]. The report provides an overview of European countries' efforts to leverage and scale up digital health to strengthen health systems and enable care provision (Ibid.).

This paper aims to present, synthesize, and interpret results from the Telehealth Programmes section of the 2022 WHO/Europe DH Survey while using Norwegian telehealth experiences to illustrate survey findings. Previous WHO surveys, such as the global survey on eHealth 2015, explored the state of telehealth adoption; however, this study provides an updated and focused analysis of telehealth development in the WHO European Region following the COVID-19 pandemic. By examining the implementation and use of key telehealth services, identifying barriers and facilitators, and discussing monitoring and evaluation practices, this study contributes to a better understanding of the current telehealth landscape and informs future policy and practice in the region.

2. Materials and methods

2.1. Survey participants

In 2022, the WHO Regional Office for Europe invited its 53 Member States to participate in a digital health survey by nominating a national survey coordinator to identify relevant national digital health experts and solicit their input.

2.2. Survey design and data collection

The survey tool was based on the 2015 WHO Global eHealth Survey. The 2015 survey was modified to reflect recent progress and policy priorities in digital health. The survey was divided into three main sections with 95 questions. Twelve subsections focused on a specific digital health area, including telehealth programmes. WHO Regional Office for Europe launched the survey online in April 2022, allowing one survey per country to be submitted until October 2022. The survey was sent to the contacts nominated by Member States. Responses were voluntary; respondents could skip specific questions and were given the option to answer "do not know" or "not applicable." The survey was primarily implemented in digital format but was also made available in paper format, when requested. The survey instructions and questions were available in Russian and English. For telehealth, Member States were asked to provide an overview of the maturity of the services offered in their country and classify their telehealth programmes as informal (early adoption of telehealth but with no formal processes or policies available), pilot (telehealth is tested and evaluated in specific situations), or established (telehealth programmes have been running for at least two years and are expected to continue for at least another two years). All 53 countries of the WHO European Region participated in the survey.

2.3. Data analysis

The data from the 2022 survey were analysed by staff at the WHO Regional Office for Europe and by WHO/Europe consultants. Data analysis was performed in Microsoft Excel and SPSS Statistics formats. Percentages of all responses to a specific question and the number of countries giving a particular response were calculated and provided. In the analyses, "not applicable," missing answers, and "do not know" were excluded. Unless otherwise stated, use of the terms "country" or "countries" refers to one or more of the 53 Member States in the WHO European Region. Results from all countries were included.

3. Results

The data analysis shows that teleradiology, telepsychiatry, and telemedicine are the most established telehealth services used in the WHO European Region as of 2022, as shown in Fig. 1. In addition, several countries reported that the COVID-19 pandemic exposed the need for new strategies and legislation to support telehealth. The survey results highlighted in this study are presented in the following sections: (1) telehealth strategies and financing, (2) telehealth programmes and services offered by Member States of the WHO European Region, (3) barriers to implementing telehealth services, and (4) monitoring and evaluation of telehealth.

3.1. Telehealth strategies and financing

Prior to the COVID-19 pandemic, telehealth operated in many countries without the necessary legal or regulatory frameworks in place. The pandemic therefore necessitated actions to advance the legislative and regulatory landscapes for telehealth, often in very short periods of time. Of the member countries, 59 % (30 out of 51) issued new legislation, strategy, policy, or guidance to support telehealth. In all, 78 % of countries (40 out of 51) stated that their country had either a national telehealth strategy or that telehealth was part of a broader digital health strategy. However, the scope of the regulatory and policy changes varied significantly. While some countries introduced narrow and targeted legislation specifically for COVID-19 patient monitoring and treatment, others introduced national strategies or laws to govern the use of

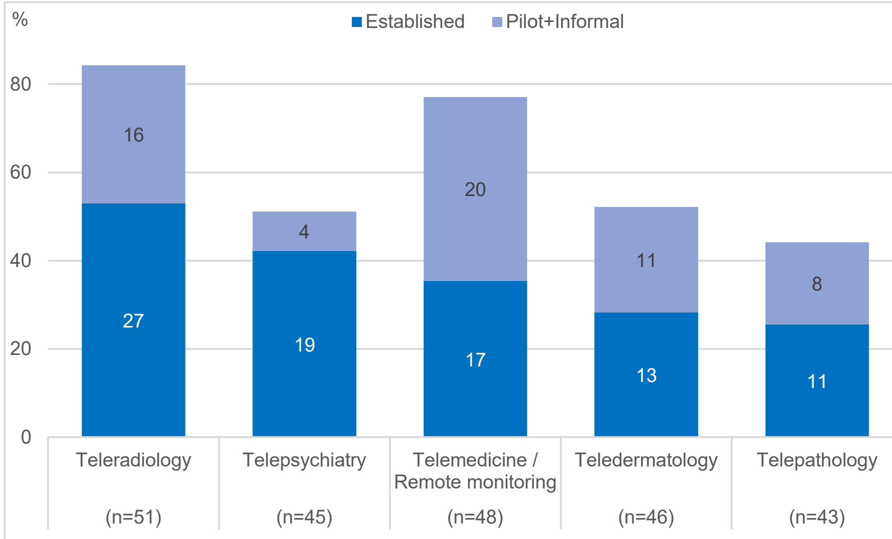

Fig. 1. Maturity of telehealth services offered by Member States of the WHO European Region *Note: "n=" is the number of respondents to each question on which percentage values are computed.*

telehealth technologies in general. According to the survey data, differing degrees of government funding and coverage exist for these services. Specifically, telepsychiatry is covered by government or compulsory financing schemes in 82 % of countries (18 out of 22). Teleradiology is covered in 74 % of the countries (28 out of 38), telemedicine in 73 % (24 out of 33), telepathology in 69 % (11 out of 16 responding countries), and teledermatology in 65 % (13 out of 20).

3.2. Telehealth programmes and services offered by Member States of the WHO European Region

3.2.1. Teleradiology, telepsychiatry and telemedicine

Teleradiology was reported by most Member States, with 84 % (43 of 51 countries) reporting presence of the service (Fig. 1.1). Of 51 Member States responding to the question about the maturity of the service, 53 % (27 countries) indicated they had established teleradiology services. Another 31 % (16 countries) were running pilot projects or offering teleradiology as informal services.

Telepsychiatry and telemedicine are the next most popular established telehealth services (Fig. 1). Just over half of the countries (23 out of 45, 51 %) offer telepsychiatry. Additionally, 42 % (19 out of 45) have fully established telepsychiatry services, while 9 % (4) have indicated that the telepsychiatry services are either in a pilot stage or being offered informally. Telemedicine and remote patient monitoring services are offered by 77 % of countries (37 out of 48 responding countries), of which 35 % (17 countries) reported having established telemedicine services, with another 42 % (20 countries) in the early phases of piloting or offering these services informally. These findings suggest that teleradiology, telepsychiatry, and telemedicine are the most widely adopted and mature telehealth services in the WHO European Region based on the self-reported data from the Member States.

Among all the services offered, telemedicine was the most introduced and improved service during the pandemic. (Fig. 2). Of the 35 responding countries, 26 % (9 countries) reported having introduced

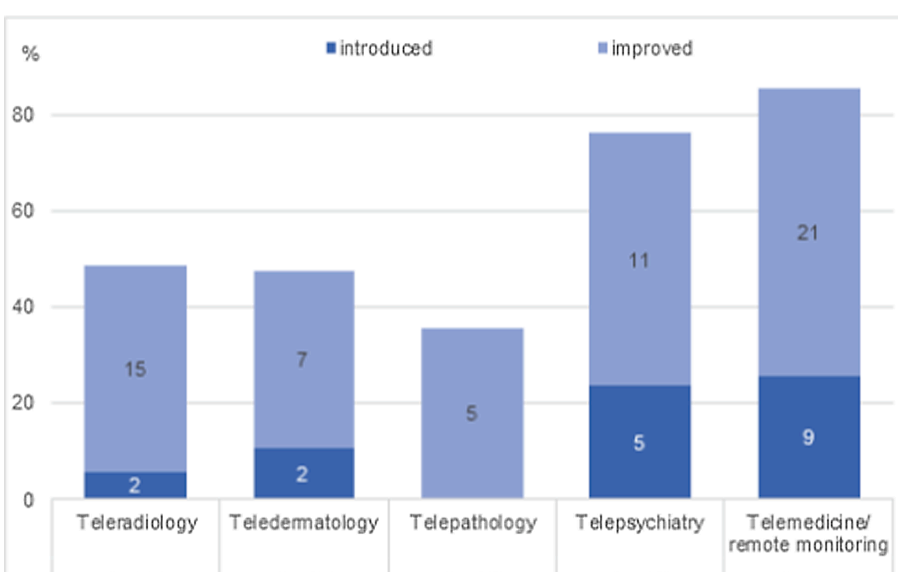

Fig. 2. Telehealth services introduced or improved during the covid pandemic. *Note: "n=" is the number of respondents to each question on which percentage values are computed.*

telemedicine services during the pandemic. More than half of countries (21, 60 %) said that their telemedicine services improved because of the pandemic. Telepsychiatry was also substantially boosted by the pandemic. Of the responding countries, 52 % (11 out of 21) reported improvement in their telepsychiatry services (Fig. 2), while 24 % of countries (5 out of 21) reported that telepsychiatry services had been introduced because of the pandemic.

3.2.2. Teledermatology and telepathology

Teledermatology, or the remote diagnosis and treatment of skin disorders, has a mixed presence across surveyed countries (Fig. 1). While it has gained traction in about half of the countries (24 out of 46), only 28 % [13] recognise it as a fully established service. Telepathology, where 44 % report its use (19 out of 43), has also experienced varied adoption rates, with only 26 % [11] indicating established telepathology services.

3.2.3. Additional telehealth services

Countries were asked to provide information about additional telehealth services for diagnosis, consultation, rehabilitation, distance learning, or other interventions. Twenty-seven Member States reported other telehealth initiatives ranging from teleconsultation, telerehabilitation to telesurgery. Teleconsultation was the most cited "other" initiative (9 countries), followed by distance learning (5 countries).

3.3. Barriers to implementing telehealth services

Survey respondents were asked to rate predefined barriers according to how important they were perceived in their country. Funding, infrastructure, and capacity/human resources were most often cited as moderate to extremely important barriers. Many countries (39 out of 51, 77 %) identified funding as their most important barrier. (Fig. 3). Funding was also cited by European respondents in the 2015 WHO Global eHealth survey as their most important barrier, across all dimensions surveyed [17]. This suggests that while the COVID-19 pandemic may have resulted in a temporary increase of funding for digital health, long-term efforts to scale up services require increased political commitment to ensure sustained funding for digital health. Legal issues, effectiveness, demand, and policy barriers were generally considered of lesser importance.

3.4. Monitoring and evaluation of telehealth

Survey responses indicate that telehealth programmes are not being evaluated systematically. Only 37 % (16 out of 43) reported that a telehealth service in their country had been evaluated. Surprisingly, just over half of the countries reported that they do not conduct evaluations of their telehealth programmes. Furthermore, only seven countries reported that they had carried out internal or external evaluation of any telehealth service or programme employed during the pandemic.

4. Discussion

The main purpose of this paper is to provide a snapshot of the status of telehealth in the WHO European Region in 2022, drawing from responses to the 2022 WHO/Europe DH survey. Analysis of the data shows that teleradiology, telepsychiatry, and telemedicine are the most established telehealth services used in the WHO European Region. Changes were also seen in telehealth funding and regulation in most Member States.

While the pandemic boosted telehealth adoption, questions remain regarding the degree and in which contexts telehealth use will continue. Survey data shows that funding remains a major barrier to implementing of telehealth services. Several countries also took steps to solidify their investments in telehealth and establish regulatory environments to support remote care delivery. In comparison, barriers related to legal aspects, effectiveness, demand, and policy for telehealth are viewed as less critical.

Together, these actions indicate that the environment that enables telehealth use in many European Member States is maturing. While barriers to its use at scale persist, the increase in countries establishing the top three telehealth services shows notable progress.

4.1. Telehealth strategies and financing

While several countries have yet to establish regulations governing telehealth, many enacted emergency measures during the pandemic, initially introduced as temporary responses. These measures allowed the swift deployment and utilization of telehealth services, but highlighting the need for comprehensive and permanent policies [18]. In Norway, the temporary regulation allowing assessment of sick leave using e-consultations rather than face-to-face consultations was made permanent as of July 2023 [19]. Analyses from the Norwegian Centre for E-health Research showed that e-consultations for this specific use led to more efficiency and flexibility, which contributed to the policy change [8].

The pandemic also accelerated action towards payment parity for e-consultations that deliver primary healthcare services; emergency measures for reimbursement turned into permanent policies within

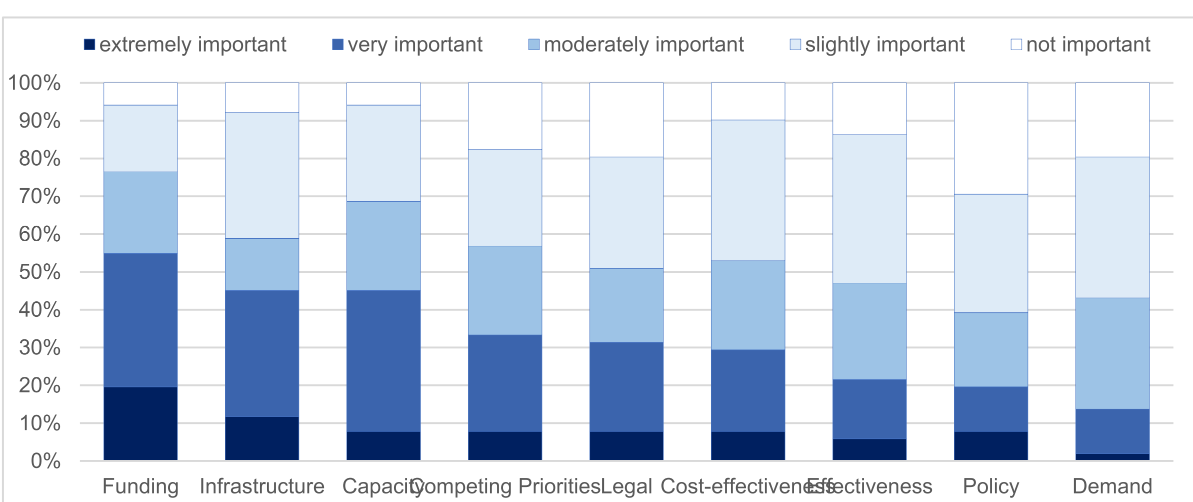

Fig. 3. Barriers to implementing telehealth initiatives.

months [20]. Since the pandemic began, specialist care services in the four health regions in Norway accelerated their investment and messaging to citizens, highlighting that digital contact with the health system is a good alternative to physical attendance at a clinic [21,22].

Funding is consistently reported as the primary barrier to telehealth adoption, and thus new funding models should be considered that support cross-organisational collaboration and avoid inequities of costs and benefits [23,24]. Prior to and following the pandemic, several countries used both the budget and fee-for-service models, while others switched to government or compulsory schemes [25,26]. In Norway, payment parity has been achieved for text, telephone, video, and physical consultations in hospitals and with general practitioners (GPs) [27]. Notably, a lack of evidence of both clinical- and cost-effectiveness is reported as a moderate or important barrier by the respondents, which detracts from further investments in new programmes [28].

4.2. Monitoring and evaluation of telehealth

Evaluation is crucial in determining if telehealth solutions are safe, cost-effective, and appropriate for local context and needs. Despite the need for evidence of their effectiveness, the survey shows that few countries carry out systematic monitoring and evaluation of telehealth programmes. Common challenges in evaluating digital health solutions include failure to include systematic monitoring and evaluation as part of the overall governance and accountability model and difficulties in identifying the necessary time, resources, and competence to evaluate effectively. Additional complexities arise if reporting the evaluation outcomes through different frameworks and to multiple stakeholders. A lack of evaluation can also create issues with funding allocation, as there is little to no evidence to guide the decision-making process [29]. Healthcare technologies cannot be successfully implemented in isolation from the wider context in which they are being used. As such, the most suitable methods of measurement depend largely on the services being evaluated and the environment in which they operate. Research shows a need to expand from a mainly technical evaluation of digital health solutions to a broader approach that includes social and organizational aspects [30,31].

In Norway, the increase in remote care resulted from the Norwegian Directorate of Health's objective to scale up digital home monitoring, underlining the advantages of a digital first approach, especially for patients with chronic conditions and multimorbidities [32]. A randomised controlled trial was conducted from 2018 to 2021 in selected regions (ibid.) to find out if remote care resulted in improved user experience and/or better health outcomes. The findings indicated that follow-up through telehealth can lead to greater patient safety and self-management. One year after implementation, mortality was lower in the intervention group, and clinicians reported that acute hospital readmissions were reduced. However, some ambiguity existed regarding cost savings, due to increased expenses associated with operating the service and the increased time the patients spent using the new solution [32].

4.3. Telehealth programmes and services offered by Member States of the WHO European Region

4.3.1. Telemedicine and telepsychiatry

The survey results show that during the pandemic, telemedicine and telepsychiatry were the services with the highest increase in adoption in the WHO European region. This has also been the case in Norway, where digital mental health services became more popular [33,34]. Several initiatives are ongoing, and video consultations, supported online-treatment, text-based communication, and chatbots are among the most commonly deployed technologies. The health authorities in Norway aim to further increase their use [35]. One example is eMeistring, a platform for supervised or supported online treatment for people with mental health disorders. Supervised online treatment includes the same elements as traditional cognitive behavioural therapy (CBT). The therapist uses evidence-based interventions, and the treatment is modular, lasting up to 14 weeks [36].

4.3.2. Teleradiology

Teleradiology has been pioneered in telehealth, providing remote access to digital radiological images and enabling specialists to deliver prompt and accurate diagnoses from a distance. In the 2009 and 2015 WHO Global surveys, teleradiology was the most prevalent service in the WHO European Region [37,38], and it was also the most established service in the 2022 survey. In Norway, teleradiology has been well-established for three decades [39]. In recent years, the use of AI for image analysis has emerged as a promising solution to increase the quality of radiology services and reduce analysis time [40,41].

An ongoing project in Norwegian radiology departments examining AI use in image analysis is expected to provide several benefits, including increased quality through more precise and efficient diagnostics and treatment, better planning, increased opportunities for research and development work, and increased sustainability of healthcare services over the long-term [40].

4.3.3. Additional telehealth services

Other telehealth services cited by respondents include teleconsultation (9 countries) followed by distance learning and telerehabilitation. These service types cover a range of specialties and illustrate the potential of telehealth to be applied across all areas of health care. However, individual survey respondents may define these terms differently.

A recent study from Norway, Denmark, and Australia compared the standard treatment of chronic obstructive pulmonary disease (COPD) to telerehabilitation and unsupervised training at home [42]. The results showed that both telerehabilitation and unsupervised training can reduce hospital readmissions. Unsupervised training is a cost-effective intervention but may not be suitable for all patients, as it requires a high degree of self-management. Telerehabilitation could therefore be a viable alternative for this patient group and for patients living in remote areas.

4.3.4. The way forward for telehealth – Addressing barriers and facilitators

The COVID-19 pandemic proved that the rapid scaling and implementation of telehealth is possible, given the right incentives. Further national efforts to leverage insights and experiences regarding telehealth gained during the pandemic require systematic assessment and application of lessons learned to help incorporate telehealth into routine care. Without adequate monitoring and evaluation mechanisms however, this may prove difficult. One research group took a closer look at Denmark, Norway, the UK, the Netherlands, and Germany and identified different post-pandemic barriers in general practice that delay or make the use of video consultations difficult [43]. They identify inadequate technological and financial support for GPs, cultural elements like the professional identity of doctors, and clinical norms as impacting the adoption of video consultations. The researchers call for a closer collaboration among all stakeholders in the implementation process to better understand and reflect the reality of general practice settings, rather than relying on policy optimism. Findings from several studies and systematic reviews indicate that issues around implementation of telehealth are complex and interrelated, and that the same or similar factors are frequently reported [44–46]. There is no single factor that determines the success or failure of an initiative [47–49].

Countries should consider actively investigating how telehealth can contribute to achieving universal health coverage. Telehealth has an immense potential to address gaps in healthcare access and improve overall quality of care, but its value in this context needs to be captured as part of a systematic evaluation of telehealth services. By utilizing telehealth services, countries can work towards ensuring equitable access to care, ultimately improving the health and well-being of their populations.

Other important applications of telehealth that contribute towards universal health coverage include improving digital literacy and capacity building among healthcare workers and individuals [50]. Reducing poverty and increasing education are among the factors that can increase digital literacy [51]. Furthermore, policies need to be developed that grant affordable and universal access to telehealth solutions. Better access to the internet will also promote digital inclusion [52].

5. Conclusion

Based on WHO's 2022 survey, the use of telehealth in the WHO European Region is on the rise. However, merely having telehealth in place is not sufficient for its successful and sustained use as a method of care provision. The analysis confirms that a systems approach is needed to achieve digital transformation in the health sector. Key digital health challenges such as financing and reimbursement, privacy and security, interoperability, digital literacy, governance, and the need for new and more effective partnerships must be addressed for telehealth to scale in line with the strategic objectives put forth by the Regional digital health action plan for the WHO European Region 2023–2030 [53].

Member States should have clear telehealth policies and strategies in place to guide its integration into the healthcare system and to foster different modes of collaboration between health system entities and across health and social sectors more broadly. The monitoring and evaluation of telehealth is essential to inform service design, policy, and practice, taking local contexts into account. Such evaluation ensures the quality of telehealth use and guides current service operation. It also provides incentive for future investment and allows assessment of the role of telehealth in working towards universal health coverage. Finally, Member States should consider strengthening current funding and reimbursement models for telehealth to improve their ability to scale up.

5.1. Limitations of the study

The main limitations of the survey relate to the fluidity of some of the terms used in the questions and specifically how these terms, many of which were defined according to the 2015 survey and standard WHO definitions, are interpreted in each of the participating countries. This applies to, for example, questions about electronic health record systems and mHealth. Additionally, non-responses to some questions hampered the robustness of the interpretations and comparisons that can be made from the survey data. Although this is difficult to avoid, making definitive answers mandatory in future surveys might encourage respondents to invest sufficient time and effort to gather the requested data. It is also important to note that telenursing was not specifically addressed in the WHO survey, which focused on five main telehealth services. Future research should consider including telenursing and other relevant telehealth services to provide a more comprehensive understanding of the telehealth landscape in the WHO European Region and beyond. Although this study focuses on the WHO European Region, the findings and the Norwegian experience can offer valuable insights for other countries seeking to implement or enhance telehealth services. The identified barriers, facilitators, and best practices can inform policy and decision-making processes, helping countries adapt their strategies to their specific contexts. Furthermore, the study highlights the importance of monitoring and evaluating telehealth interventions, which is crucial for ensuring their effectiveness, efficiency, and sustainability, regardless of the country or region.

6. Disclaimer

The authors affiliated with the World Health Organization (WHO) are alone responsible for the views expressed in this publication and they do not necessarily represent the decisions or policies of the WHO.

7. Authors statement

All the authors have reviewed and approved the final manuscript.

CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We acknowledge the support and assistance provided by colleagues at the Norwegian Centre for E-health Research, especially Trine Bergmo for her revision and comments on the content and staff of the Data and Digital Health Programme at the WHO Regional Office for Europe.

References

[1] A. Buvik, T.S. Bergmo, E. Bugge, A. Smaabrekke, T. Wilsgaard, J.A. Olsen, Cost-effectiveness of telemedicine in remote orthopedic consultations: randomized controlled trial, J. Med. Internet Res. 21 (2) (2019) e11330.
[2] P. Berggren, E. Pourazar, A. Lundqvist, Access to Rural Services by Strengthening Primary Care with Digital Tools in Remote Areas of Sweden, World Health Organization, Country Vignette, 2022.
[3] The ongoing journey to commitment and transformation Digital health in the WHO European Region: World Health Organization - Regional Office for Europe, 2023. 50 p.
[4] Global diffusion of eHealth: making universal health coverage achievable: report of the third global survey on eHealth. World Health Organization, 2016.
[5] Centers for Disease Control and Prevention. Telehealth and Telemedicine: A Research Anthology of Law and Policy Resources USA: Centers for Disease Control and Prevention, 2023 <https://www.cdc.gov/phlp/publications/topic/anthologies/anthologies-telehealth.html> (cited 2023 27.06).
[6] World Health Organization, Telehealth: Analysis of Third Global Survey on eHealth Based on the Reported Data by Countries, WHO, Geneva, 2016.
[7] World Health Organization, Consolidated telemedicine implementation guide, WHO, 2022.
[8] E. Kristiansen, E. Breivik, T.S. Bergmo, M. Johansen, P. Zanaboni, E-konsultasjon og sykmelding. Undersøkelse av erfaringer med unntak fra krav til personlig fremmøte under covid-19-pandemien, Norwegian Centre for E-health Research, Tromsø, Norway, 2021.
[9] M.K. Gullslett, E. Kristiansen, E.R. Nilsen, Therapists' experience of video consultation in specialized mental health services during the COVID-19 pandemic: qualitative interview study, JMIR Hum. Fact. 8 (3) (2021) e23150.
[10] Telehealth: National Institute of Biomedical Imaging and Bioengineering, 2020. <https://www.nibib.nih.gov/science-education/science-topics/telehealth>.
[11] K. Aristodemou, L. Buchhass, D. Claringbould, The COVID-19 crisis in the EU: the resilience of healthcare systems, government responses and their socio-economic effects, Eurasian Econ. Rev. 11 (2021) 251–281.
[12] S. Doraiswamy, A. Abraham, R. Mamtani, S. Cheema, Use of telehealth during the COVID-19 pandemic: scoping review, J. Med. Internet Res. 22 (12) (2020) e24087.

[13] World Health Organization, Statement on the fifteenth meeting of the IHR (2005) Emergency Committee on the COVID-19 pandemic: WHO, 2023 (cited 2023 11.05.2023). <https://www.who.int/news/item/05-05-2023-statement-on-the-fifteenth-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-coronavirus-disease-(covid-19)-pandemic>.

[14] A.K.C. Wong, J. Bayuo, F.K.Y. Wong, V.W.Y. Kwok, D.W.K. Tong, M.K. Kwong, et al., Sustaining telecare consultations in nurse-led clinics: Perceptions of stroke patients and advanced practice nurses: a qualitative study, Digit. Health. 9 (2023).

[15] A.Y.L. Lee, A.K.C. Wong, T.T.M. Hung, J. Yan, S. Yang, Nurse-led telehealth intervention for rehabilitation (telerehabilitation) among community-dwelling patients with chronic diseases: systematic review and meta-analysis, J. Med. Internet Res. 24 (11) (2022) e40364.

[16] E.R. Nilsen, J. Dugstad, H. Eide, M.K. Gullslett, T. Eide, Exploring resistance to implementation of welfare technology in municipal healthcare services – a longitudinal case study, BMC Health Serv. Res. 16 (1) (2016) 657.

[17] From innovation to implementation: eHealth in the WHO European Region. World Health Organization. Regional Office for Europe, 2016.

[18] Ardielli E. eHealth in the European Union–Comparative Study. Science of Economics. 2020:7.

[19] nav.no. Sykmelding ved e-konsultasjon – endring i folketrygdloven fra 1. juli 2023, 2023 <https://www.nav.no/no/nav-og-samfunn/samarbeid/leger-og-andre-behandlere/nyheter/sykmelding-ved-e-konsultasjon–endring-i-folketrygdloven-fra-1.juli-2023>.

[20] Helsenorge.no. User fees at the family doctor, 2023 <https://www.helsenorge.no/en/payment-for-health-services/user-fees-at-the-family-doctor/>.

[21] Northern Norway Regional Health Authority. Videokonsultasjon <https://unn.no/behandlinger/videokonsultasjon>.

[22] The Western Norway Regional Health Authority. Vil ha fleire telefon- og videokonsultasjonar i sjukehusa, 2023. <https://www.helse-vest.no/nyheiter/nyheiter-2022/vil-ha-fleire-telefon-og-videokonsultasjonar-i-sjukehusa/>.

[23] H. Alami, S.E. Shaw, J.-P. Fortin, M. Savoldelli, R. Fleet, B. Têtu, The 'wrong pocket'problem as a barrier to the integration of telehealth in health organisations and systems, Dig. Health. 9 (2023).

[24] L. Garattini, M. Badinella Martini, P.M. Mannucci, Improving primary care in Europe beyond COVID-19: from telemedicine to organizational reforms, Intern. Emerg. Med. 16 (2) (2021) 255–258.

[25] R. Fisher, U. Sarkar, J. Adler-Milstein, Audio-Only Telemedicine In Primary Care: Embraced In The NHS, Second Rate In The US. Health Affairs Forefront. 2022.

[26] P. Kouri, J. Reponen, O. Ahonen, P. Metsäniemi ,A. Holopainen, E. Kontio, Telemedicine and eHealth in Finland: on the way to digitalization–from individual telehealth applications to connected health. A Century of Telemedicine: Curatio Sine Distantia et Tempora A World Wide Overview–Part II, 2018.

[27] The Norwegian Directorate of Ehealth, National e-Health Strategy for the Health and Care Sector 2023–2030, The Norwegian Directorate of E-health, Oslo, 2023.

[28] N.D. Eze, C. Mateus, C.O. Hashiguchi, T., Telemedicine in the OECD: an umbrella review of clinical and cost-effectiveness, patient experience and implementation, PLoS One 15 (8) (2020) e0237585.

[29] K. Kidholm, J. Clemensen, L.J. Caffery, A.C. Smith, The Model for Assessment of Telemedicine (MAST): a scoping review of empirical studies, J. Telemed. Telecare 23 (9) (2017) 803–813.

[30] C. Jacob, J. Lindeque, A. Klein, C. Ivory, S. Heuss, M.K. Peter, Assessing the quality and impact of ehealth tools: systematic literature review and narrative synthesis, JMIR Hum. Fact. 10 (2023) e45143.

[31] J. Mielke, S. De Geest, F. Zúñiga, T. Brunkert, L.L. Zullig, L.M. Pfadenhauer, et al., Understanding dynamic complexity in context—enriching contextual analysis in implementation science from a constructivist perspective, Front. Health Services. 2 (2022).

[32] The Norwegian Directorate of Health, Trial of Digital Home Monitoring in Norway, The Norwegian Directorate of Health, Oslo, 2021.

[33] L.L. Bielinski, T. Berger, Internet interventions for mental health: Current state of research, lessons learned and future directions, Konsul'tativnaia Psikhol. Psikhoterapiia. 28 (3) (2020) 65–83.

[34] R. Yao, W. Zhang, R. Evans, G. Cao, T. Rui, L. Shen, Inequities in health care services caused by the adoption of digital health technologies: scoping review, J. Med. Internet Res. 24 (3) (2022) e34144.

[35] The Norwegian Ministry of Health and Care Services, Plan for increase mental health services (2023–2033), The Norwegian Ministry of Health and Care Services, Oslo, 2023.

[36] A. Yogarajah, R. Kenter, Y. Lamo, V. Kaldo, T. Nordgreen, Internet-delivered mental health treatment systems in Scandinavia–a usability evaluation, Internet Interv. 20 (2020) 100314.

[37] World Health Organization, Global Diffusion of eHealth: Making Universal Health Coverage Achievable: Report of the Third Global Survey on eHealth, World Health Organization, Geneva, 2016.

[38] World Health Organization GOfe, Telemedicine: Opportunities and Developments in Member States: Report on the Second Global Survey on eHealth, World Health Organization, Geneva, 2010.

[39] A. Stormo, S. Sollid, J. Størmer, T. Ingebrigtsen, Neurosurgical teleconsultations in northern Norway, J. Telemed. Telecare 10 (3) (2004) 135–139.

[40] L. Silsand, G.-H. Severinsen, L. Linstad, G. Ellingsen, Procurement of artificial intelligence for radiology practice, Proc. Comput. Sci. 219 (2023) 1388–1395.

[41] T. Boeken, J. Feydy, A. Lecler, P. Soyer, A. Feydy, M. Barat, et al., Artificial intelligence in diagnostic and interventional radiology: Where are we now? Diagn Interv Imag. 104 (1) (2023) 1–5.

[42] P. Zanaboni, B. Dinesen, H. Hoaas, R. Wootton, A.T. Burge, R. Philp, et al., Long-term telerehabilitation or unsupervised training at home for patients with chronic obstructive pulmonary disease: a randomized controlled trial, Am. J. Respir. Crit. Care Med. 207 (7) (2023) 865–875.

[43] E. Assing Hvidt, H. Atherton, J. Keuper, E. Kristiansen, E.C. Lüchau, B. Lønnebakke Norberg, et al., Low adoption of video consultations in post–COVID-19 general practice in northern europe: barriers to use and potential action points, J. Med. Internet Res. 25 (2023) e47173.

[44] T. Greenhalgh, H. Maylor, S. Shaw, J. Wherton, C. Papoutsi, V. Betton, et al., The NASSS-CAT tools for understanding, guiding, monitoring, and researching technology implementation projects in health and social care: protocol for an evaluation study in real-world settings, JMIR Res. Protocols. 9 (5) (2020) e16861.

[45] U.H. Tang, F. Smith U. Karilampi, A. Gremyr, Understanding the role of complexity in bottom-up healthcare innovations: a multiple case study using the NASSS-CAT tool, 2022.

[46] A.D. Van Citters, O. Dieni, P. Scalia, C. Dowd, K.A. Sabadosa, J.D. Fliege, et al., Barriers and facilitators to implementing telehealth services during the COVID-19 pandemic: a qualitative analysis of interviews with cystic fibrosis care team members, J. Cyst. Fibros. 20 (2021) 23–28.

[47] J. Greenhalgh, A. Manzano, Understanding 'context'in realist evaluation and synthesis, Int. J. Soc. Res. Methodol. 25 (5) (2022) 583–595.

[48] L. Otto, L. Harst, Investigating barriers for the implementation of telemedicine initiatives: a systematic review of reviews, in: Twenty-fifth Americas Conference on Information Systems, Cancun, 2019; Cancun. Cancun: Americas Conference on Information System, 2019.

[49] F. Saigí-Rubió, I.J. Borges do Nascimento, N. Robles, K. Ivanovska, C. Katz, N. Azzopardi-Muscat, et al., The current status of telemedicine technology use across the world health organization european region: an overview of systematic reviews, J. Med. Internet Res. 24(10) (2022).

[50] R. Van Kessel, R. Hrzic, E. O'Nuallain, E. Weir, B.L.H. Wong, M. Anderson, et al., Digital health paradox: international policy perspectives to address increased health inequalities for people living with disabilities, J. Med. Internet Res. 24 (2) (2022) e33819.

[51] G. Quaglio, K. Sørensen, P. Rübig, L. Bertinato, H. Brand, T. Karapiperis, et al., Accelerating the health literacy agenda in Europe, Health Promot Int. 32 (6) (2017) 1074–1080.

[52] A. Blandford, J. Wesson, R. Amalberti, R. AlHazme, R. Allwihan, Opportunities and challenges for telehealth within, and beyond, a pandemic, Lancet Global Health. 8 (11) (2020) e1364–e1365.

[53] World Health Organization Regional Committee for Europe ns. Regional digital health action plan for the WHO European Region 2023–2030. . Seventy-second Regional Committee for Europe: Tel Aviv, 12–14 September 2022:; 2022 12.14.9.2022. Contract No.: EUR/RC72/5 - Provisional agenda item 3.